

Provenance Data Model - Professional ERD v3

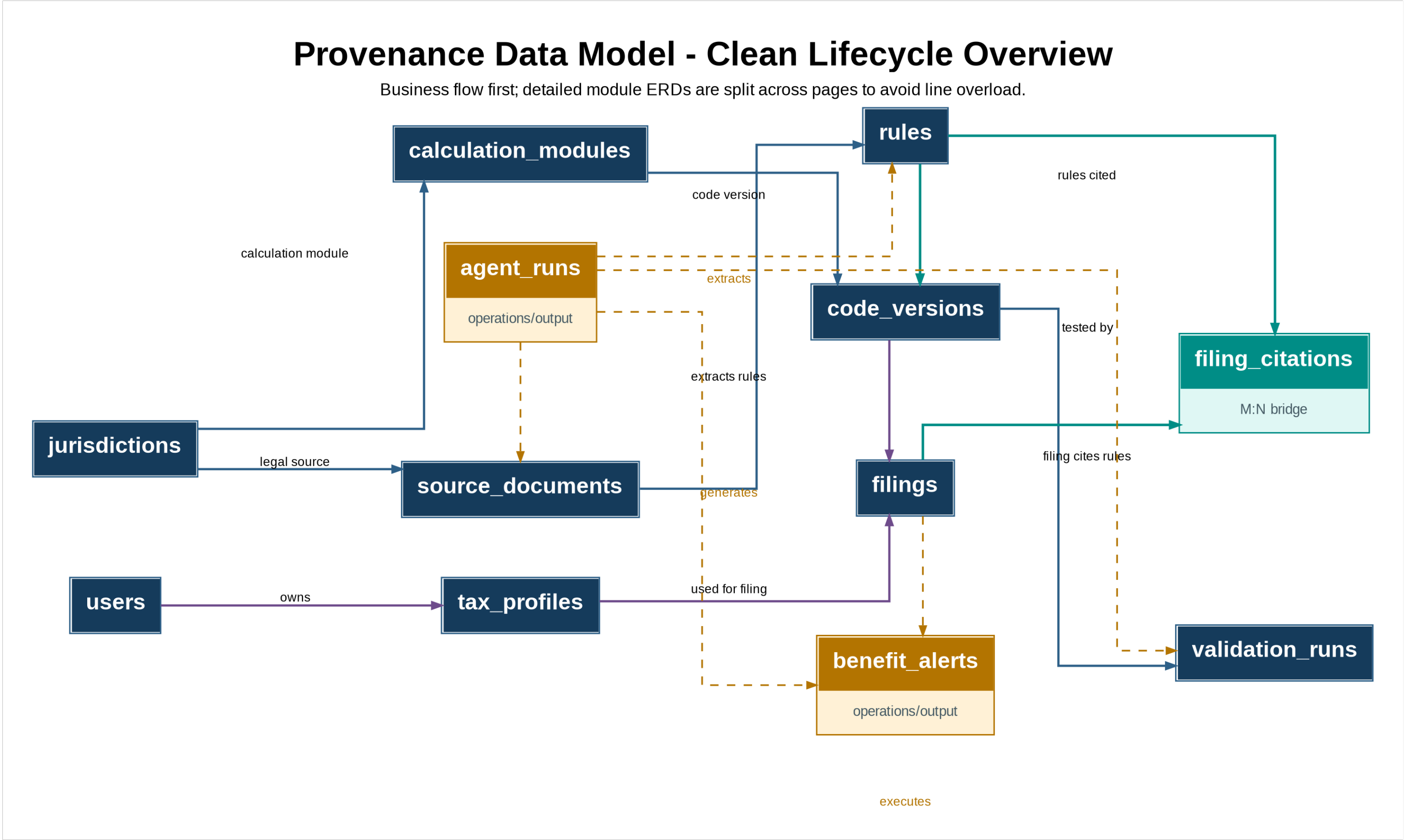
Cleaned for clarity, maintainability, and implementation. This revision separates the model into readable modules, uses consistent table naming, and turns previously unsupported relationship labels into explicit foreign keys or removes them from the core ERD.

Design decision	V3 choice	Why it matters
Naming	Plural lower_snake_case table names	Consistent names reduce implementation mistakes and improve migrations.
Diagram layout	One lifecycle overview + four detailed module ERDs	A single giant ERD is not readable for this domain.
Agent runs	Moved to operations/provenance module with explicit optional FKs	No more fake lines; every relationship has a column behind it.
Filings	filings.tax_profile_id replaces weak user_id + tax_year linkage	A filing now points to the exact profile used, preventing data drift.
Document lineage	Renamed to document_citation_index unless historical lineage is added	The existing table maps a citation key to the current document; that is not full lineage.

Hard truth: the previous ERD was structurally useful but visually overloaded. V3 treats the full schema as a documentation pack, not as one unreadable diagram.

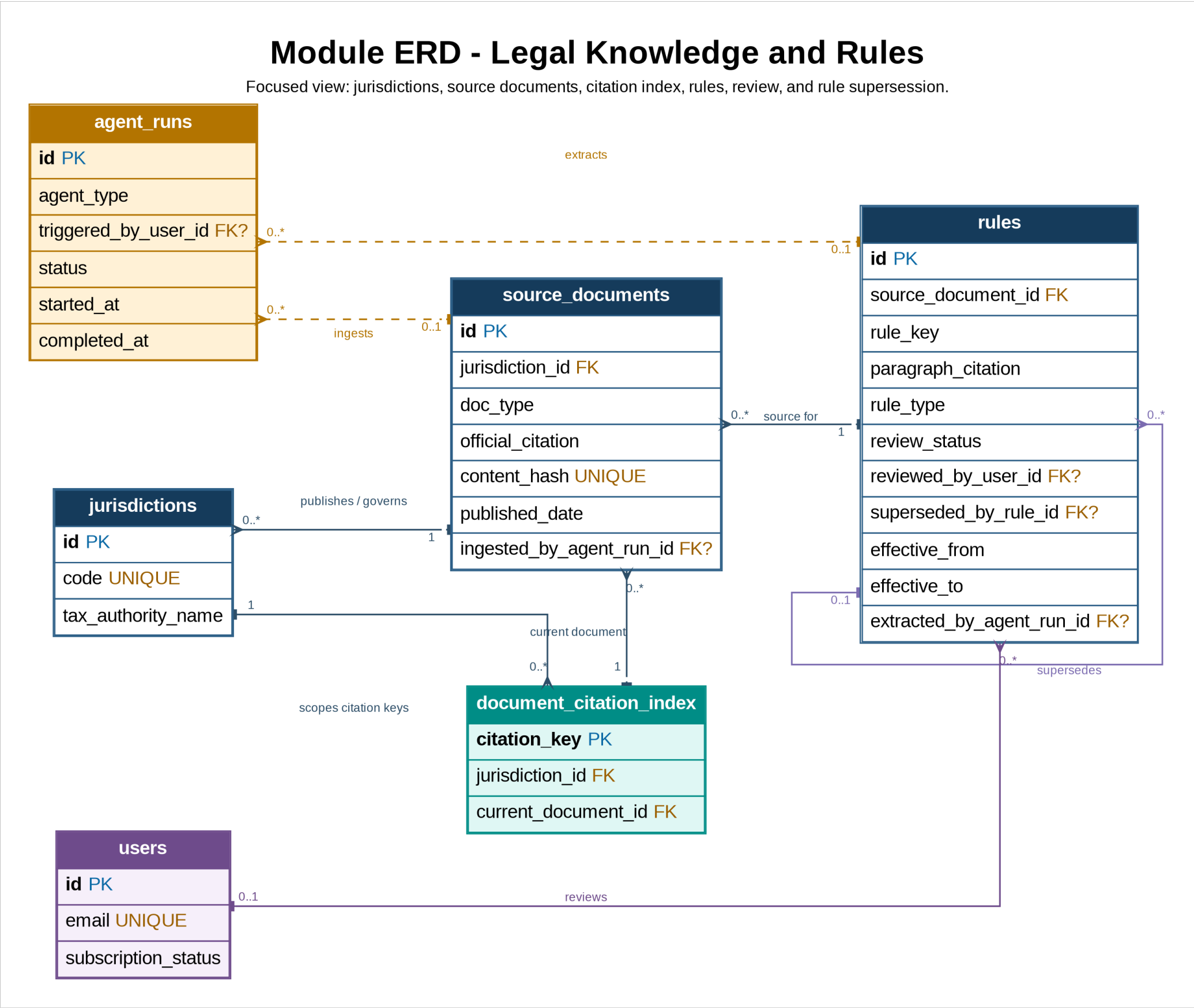
1. Lifecycle overview

This is the diagram to use first in a pitch, README, or report. It shows the full business flow without drowning the reader in columns.



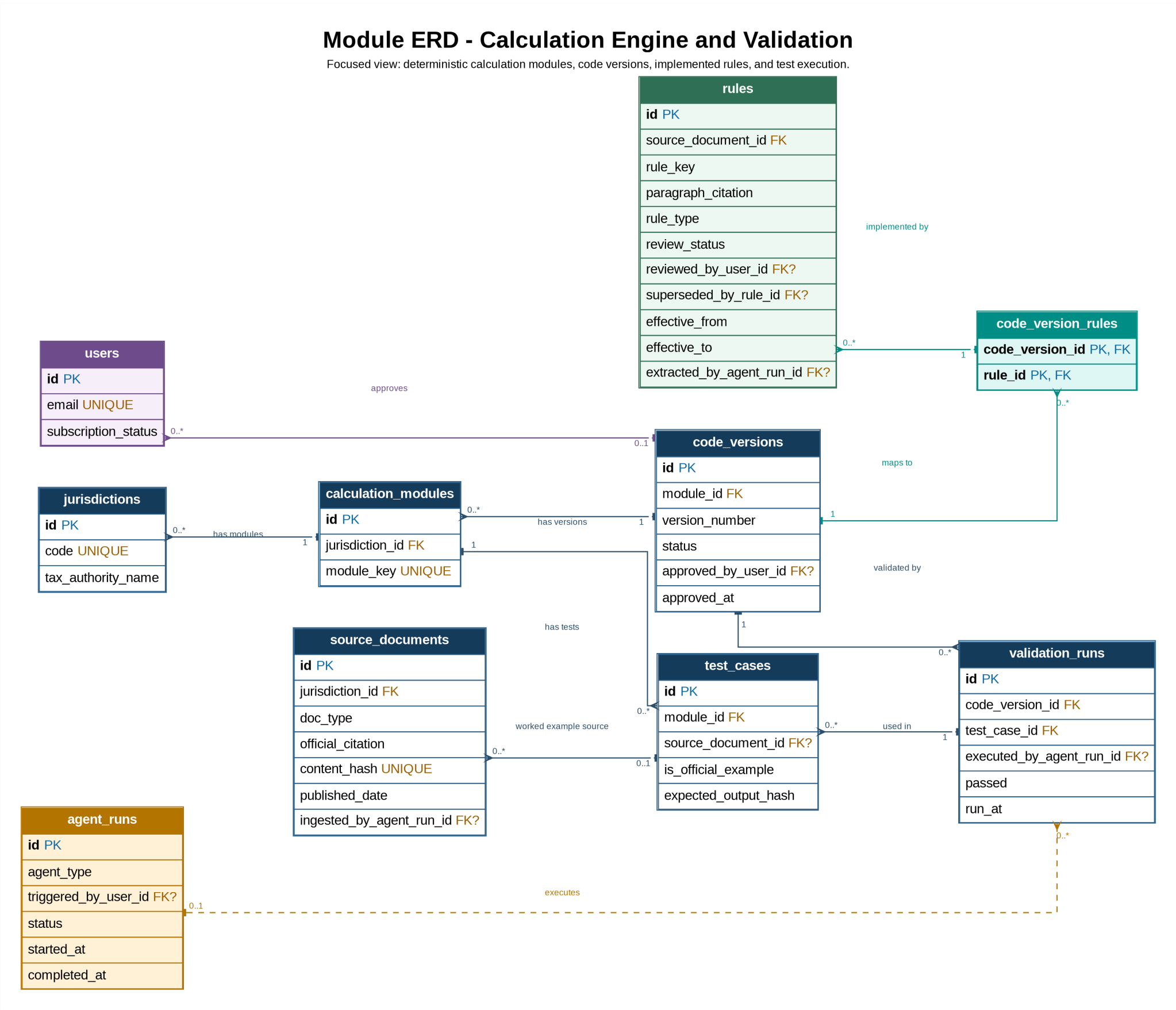
2. Legal knowledge and rules

This module shows legal source documents, citation indexing, rules, review, supersession, and agent extraction provenance.



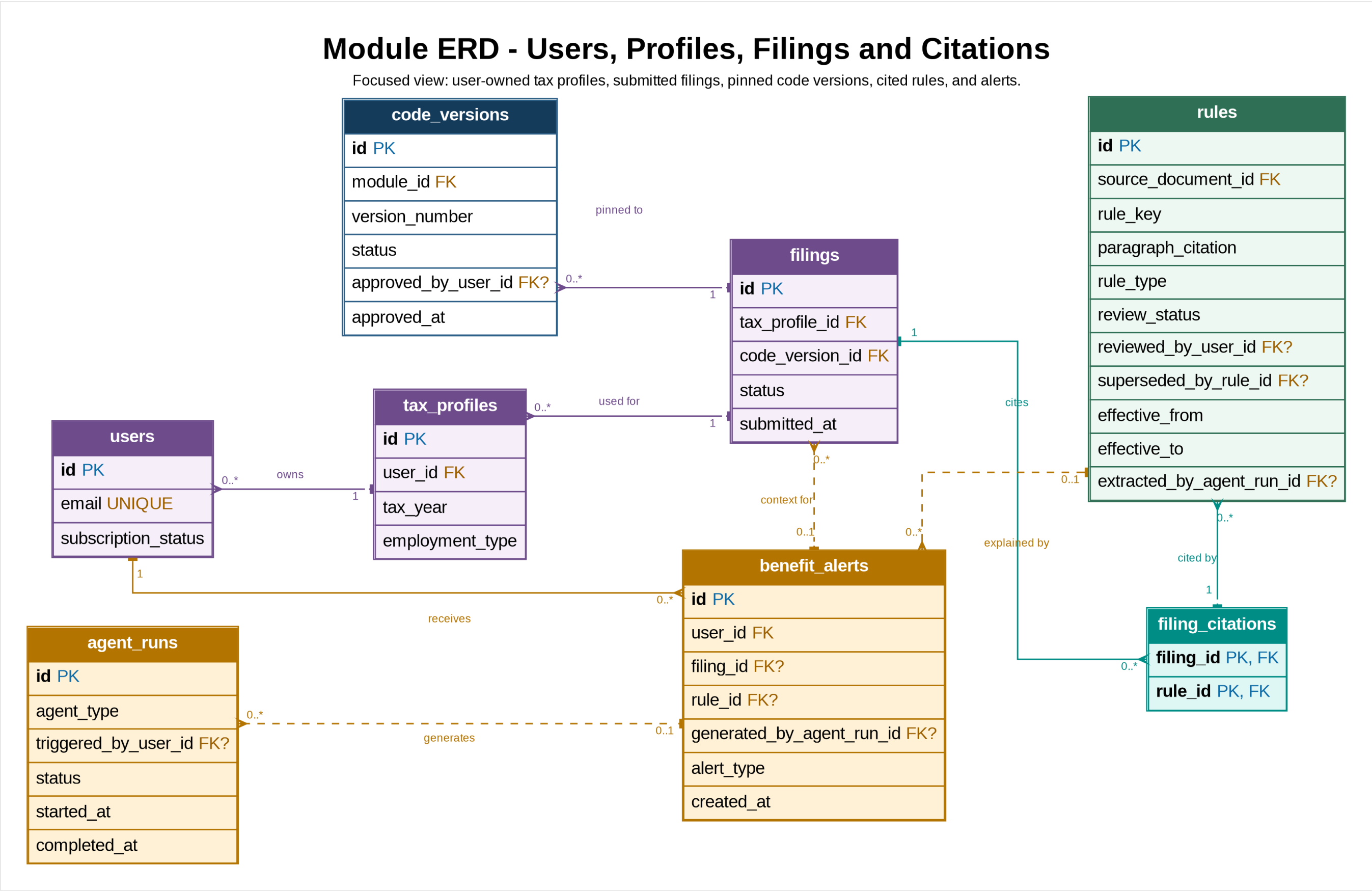
3. Calculation engine and validation

This module shows the deterministic engine: modules, code versions, implemented rules, official worked examples, and validation runs.



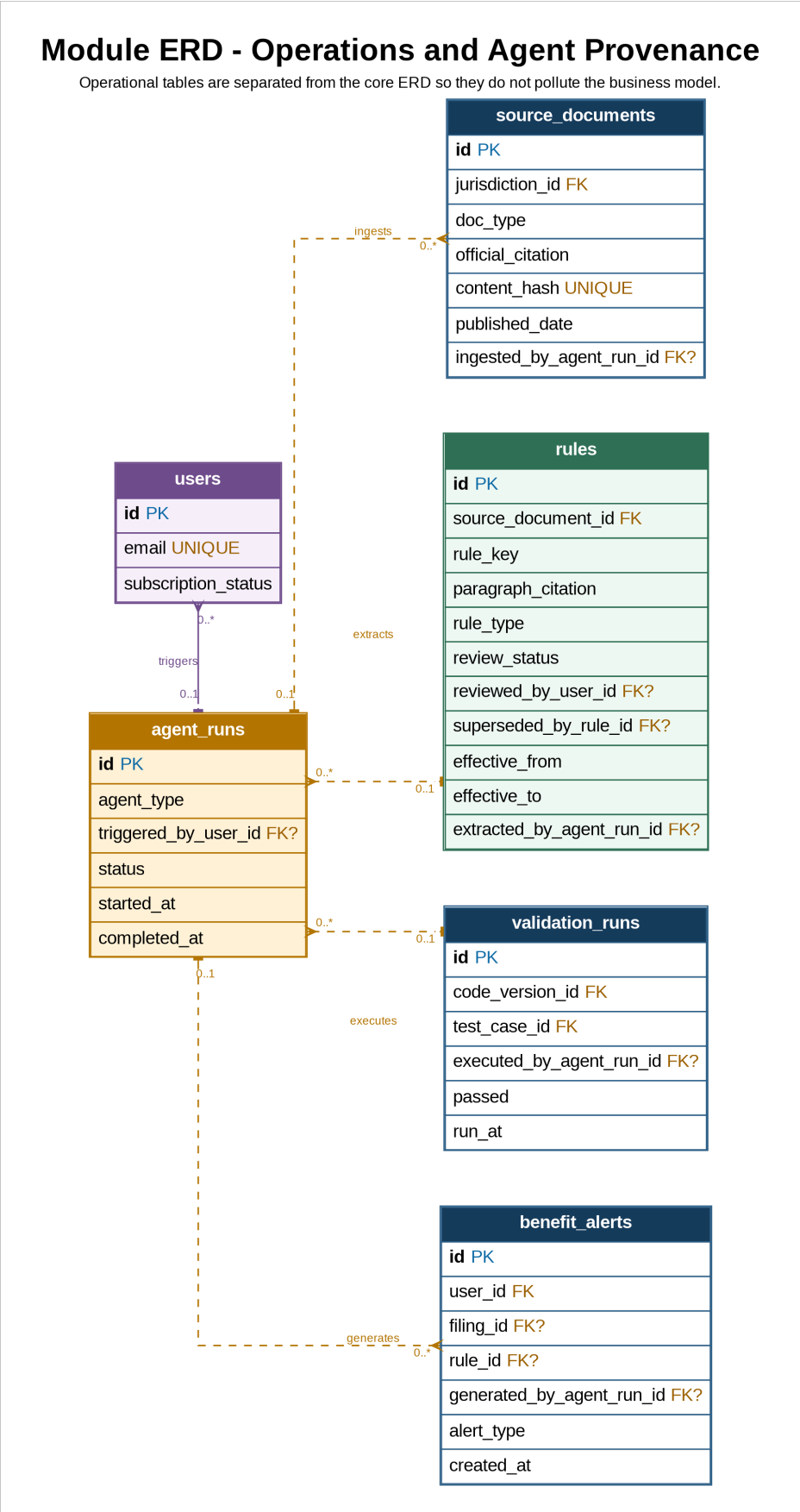
4. Users, profiles, filings and citations

This module shows how user tax profiles produce filings pinned to a code version, with rule citations for explainability.



5. Operations and agent provenance

This module keeps operational traceability separate from the core data model. That is cleaner than forcing agent-run lines into every diagram.



Relationship Catalogue

Every relationship below is represented by an explicit FK in the recommended v3 schema. Optional operational FKs are marked with nullable intent in the entity diagrams.

Parent	Child	Cardinality	Label	FK column
jurisdictions	source_documents	1 -> 0..*	publishes / governs	source_documents.jurisdiction_id
jurisdictions	calculation_modules	1 -> 0..*	has modules	calculation_modules.jurisdiction_id
jurisdictions	document_citation_index	1 -> 0..*	scopes citation keys	document_citation_index.jurisdiction_id
source_documents	rules	1 -> 0..*	is source for	rules.source_document_id
source_documents	document_citation_index	1 -> 0..*	current document	document_citation_index.current_document_id
rules	rules	0..1 -> 0..*	supersedes	rules.superseded_by_rule_id
calculation_modules	code_versions	1 -> 0..*	has versions	code_versions.module_id
calculation_modules	test_cases	1 -> 0..*	has tests	test_cases.module_id
source_documents	test_cases	0..1 -> 0..*	cites worked example	test_cases.source_document_id
code_versions	code_version_rules	1 -> 0..*	maps to	code_version_rules.code_version_id
rules	code_version_rules	1 -> 0..*	implemented by	code_version_rules.rule_id
code_versions	validation_runs	1 -> 0..*	validated by	validation_runs.code_version_id
test_cases	validation_runs	1 -> 0..*	used in	validation_runs.test_case_id
users	code_versions	0..1 -> 0..*	approves	code_versions.approved_by_user_id
users	rules	0..1 -> 0..*	reviews	rules.reviewed_by_user_id
users	tax_profiles	1 -> 0..*	owns	tax_profiles.user_id
tax_profiles	filings	1 -> 0..*	used for	filings.tax_profile_id
code_versions	filings	1 -> 0..*	pinned to	filings.code_version_id
filings	filing_citations	1 -> 0..*	cites	filing_citations.filing_id
rules	filing_citations	1 -> 0..*	cited by	filing_citations.rule_id
users	benefit_alerts	1 -> 0..*	receives	benefit_alerts.user_id
agent_runs	source_documents	0..1 -> 0..*	ingests	source_documents.ingested_by_agent_run_id
agent_runs	rules	0..1 -> 0..*	extracts	rules.extracted_by_agent_run_id
agent_runs	validation_runs	0..1 -> 0..*	executes	validation_runs.executed_by_agent_run_id
agent_runs	benefit_alerts	0..1 -> 0..*	generates	benefit_alerts.generated_by_agent_run_id
users	agent_runs	0..1 -> 0..*	triggers	agent_runs.triggered_by_user_id

Naming and Implementation Notes

Use this section when moving from diagram to database migrations. The aim is not to rename for aesthetics; it is to make the schema more predictable and less error-prone.

Current name / pattern	Recommended v3 name / pattern	Reason
RAW_DOCUMENTS	source_documents	Clearer: not raw forever; it is the canonical imported legal source.
EXTRACTED_RULES	rules	Rules are the business object; extraction is provenance metadata.
DOCUMENT_LINEAGE	document_citation_index	Current columns map citation_key to current_document_id; that is an index, not true lineage.
USER_TAX_PROFILES	tax_profiles	Shorter, still clear because user_id expresses ownership.
FILINGS.user_id + tax_year	filings.tax_profile_id	Avoid drift; filing should point to the exact profile used.

Minimal SQL patch for the current schema

```
-- Provenance ERD v3 - minimal implementation patch for the current schema
-- Keep existing table names but add the missing FKs that make the ERD truthful.

ALTER TABLE extracted_rules
  ADD COLUMN reviewed_by_user_id uuid NULL REFERENCES users(id),
  ADD COLUMN extracted_by_agent_run_id uuid NULL REFERENCES agent_runs(id);

ALTER TABLE validation_runs
  ADD COLUMN executed_by_agent_run_id uuid NULL REFERENCES agent_runs(id);

ALTER TABLE raw_documents
  ADD COLUMN ingested_by_agent_run_id uuid NULL REFERENCES agent_runs(id);

ALTER TABLE test_cases
  ADD COLUMN source_document_id uuid NULL REFERENCES raw_documents(id);

ALTER TABLE filings
  ADD COLUMN tax_profile_id uuid NULL REFERENCES user_tax_profiles(id);

ALTER TABLE benefit_alerts
  ADD COLUMN filing_id uuid NULL REFERENCES filings(id),
  ADD COLUMN rule_id uuid NULL REFERENCES extracted_rules(id),
  ADD COLUMN generated_by_agent_run_id uuid NULL REFERENCES agent_runs(id);

ALTER TABLE agent_runs
  ADD COLUMN triggered_by_user_id uuid NULL REFERENCES users(id);

-- Recommended cleanup after patching and migrating data:
-- 1. Replace filings.user_id + filings.tax_year lookups with filings.tax_profile_id.
-- 2. Rename raw_documents -> source_documents.
-- 3. Rename extracted_rules -> rules.
-- 4. Rename document_lineage -> document_citation_index unless true historical lineage is added.
-- 5. Keep agent_runs and benefit_alerts outside the core business ERD in an operations schema.
```